IGCSE 0607 Practice Questions — E1.6

The four operations

Total Questions: 9 — **Total Marks:** 47 Marks

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1 **(a)** The temperature at midnight was $-5°$C.
By noon, the temperature had risen by $14°$C.
Write down the temperature at noon.

..... **[1]**

(b) Work out the difference between $-8°$C and $17°$C.

..... **[1]**

(c) Work out the value of

$$24 - 12 \div (3 - 7)$$

..... **[2]**

2 **(a)** Work out the value of

$$15 - 3 \times (4 - (-2))$$

..... **[2]**

(b) Work out the value of

$$2\frac{1}{4} \div \frac{3}{8} - 1\frac{1}{2}$$

Give your answer in its simplest form.

..... **[3]**

3 **(a)** Calculate the exact value of

$$2\frac{2}{5} \div \left(1\frac{1}{3} - \frac{5}{6}\right)$$

Show all your working and give your answer as a mixed number in its simplest form.

..... **[4]**

(b) Work out the exact value of

$$\frac{0.45 \times 1.2}{0.06}$$

..... **[3]**

4 Solve the equation, giving your answer as a fraction in its simplest form.

$$\frac{x+\frac{1}{3}}{2-0.4} = -1\frac{1}{4}$$

$x =$ ..... **[5]**

5 A water tank has a capacity of 85.5 litres.
It is initially completely empty.
A hose pumps water into the tank at a rate of $\frac{3}{4}$ of a litre per minute.
At the same time, a leak in the tank loses water at a rate of 0.45 litres per minute.

Work out how long it takes to completely fill the tank.
Give your answer in hours and minutes.

hours ..... , minutes ..... **[4]**

6 A large container has a total capacity of 1000 litres.

(a) The container is initially $\frac{2}{5}$ full of water.
Water is pumped into the container at a constant rate of 18.5 litres per minute for 20 minutes.
Work out the volume of water in the container after these 20 minutes.

..... litres **[3]**

(b) Water then leaks from the container at a constant rate of 4.5 litres per minute.
Calculate the time taken for the container to become exactly half full.

..... minutes **[3]**

(c) Once the container is half full, the leak is fixed.
A different pump then fills the container completely in 1 hour and 20 minutes.
Calculate the average rate at which this pump fills the container.
Give your answer in litres per minute.

..... litres per minute **[3]**

7 **(a)** Work out the value of

$$12 - 3 \times (-4) + (15 \div -3)$$

..... **[2]**

(b) Work out the value of

$$\frac{2}{5} \div \frac{4}{7} - \frac{1}{10}$$

Give your answer as a fraction in its simplest form.

..... **[2]**

8 **(a)** At 06:00, the temperature in a city is $-4.5°C$.
By 14:00, the temperature has risen by $11.2°C$.
By 22:00, the temperature has dropped by $8.1°C$ from the temperature at 14:00.

Work out the temperature at 22:00.

..... **[2]**

(b) Use your calculator to work out the value of

$$\frac{4.72^2 - \sqrt[3]{18.5}}{3.1 \times (0.45 + 1.2)}$$

Give your answer correct to 3 significant figures.

..... **[2]**

9 Without using a calculator, work out

$$\frac{3\frac{2}{3} + 2\frac{1}{4} \times 1\frac{1}{3}}{4\frac{1}{2} - 1\frac{5}{6}}$$

You must show all your working and give your answer as a fraction in its simplest form.

..... **[5]**